Title: Parking Assistant Prototype

Overview: Practical parking assistance remains a high-value, real-world feature that reduces driver stress, decreases parking time and curbside congestion, and lowers minor-damage collisions in tight spaces. The goal of this project is to develop a simulator parking assistant that detects empty parking slots from camera input and plans a feasible reverse parking maneuver. The system should reliably infer which marked parking slots are free, convert detections into a simple ground-plane map and generate a kinematically feasible trajectory for a typical car model (considering geometric collision checking).

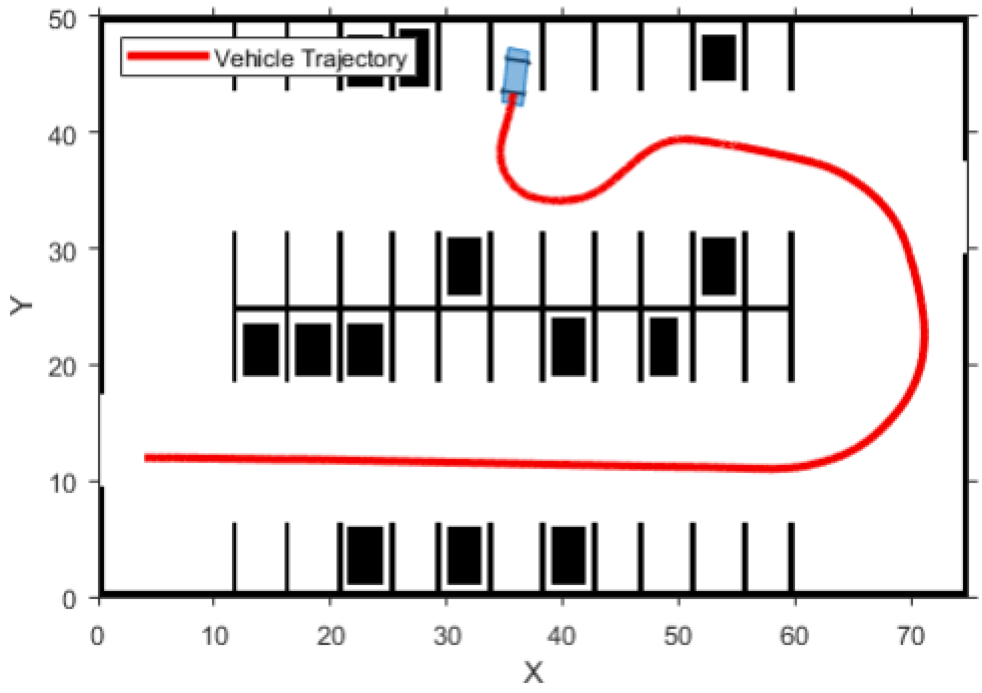
Objectives:

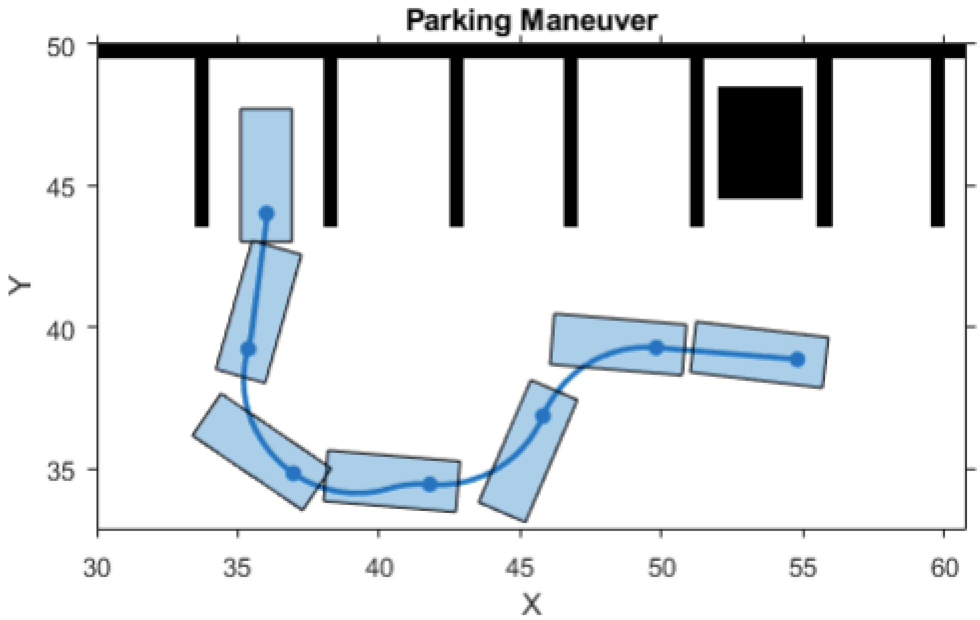
- Build a reliable and reproducible pipeline: video/image -> detection -> path planning -> execution (simulation),
- Implement a lightweight planner that considers simplified kinematic constraints,
- Develop a simple GUI for visualization,
- Demonstrate robustness and produce measured performance (success rate, detection accuracy, ...).

Expected outcomes:

- Working end-to-end demo with reproducible evaluation scripts,
- Detailed project report: scope, assumptions, architecture, algorithms, implementation, results, evaluation.

Skills to be developed: The students will learn practical skills in computer vision, geometric mapping, path planning, and measured system evaluation.


Vehicle Trajectory
Y
X
0
10
20
30
40
50
60
70


Parking Maneuver
Y
X
30
35
40
45
50
55
60